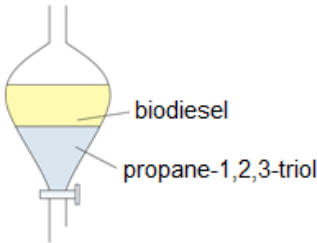
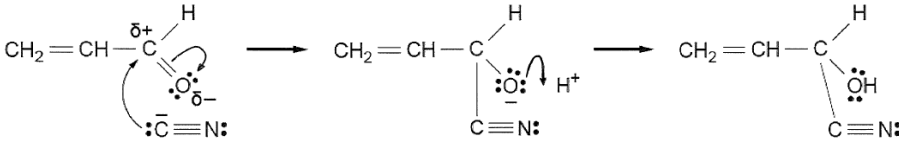


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		award (1) for either of following - the groups bonded to each carbon atom of the C=C must be the same - it must be a symmetrical alkene			1	1		
		(ii)		orange / red / yellow	1			1		1
		(iii)	I	award (1) for any of following - it does not contain a $\begin{array}{c} \text{H} \\ \diagup \\ \text{C} \\ \diagdown \\ \text{O} \end{array}$ group - it is not an aldehyde			1	1		
			II	it is not an aldehyde, therefore cannot be $\text{CH}_3(\text{CH}_2)_3\text{CHO}$ or $\text{CH}_3(\text{CH}_2)_2\text{CHO}$ (1) accept it must be $\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{CH}_3(\text{CH}_2)_3\text{C} \\ \diagdown \\ \text{O} \end{array}$ or $\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{CH}_3\text{CH}_2\text{C} \\ \diagdown \\ \text{O} \end{array}$ melting temperature cannot be higher than the literature value therefore it cannot be $\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{CH}_3(\text{CH}_2)_3\text{C} \\ \diagdown \\ \text{O} \end{array}$ (1) compound U must be $\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{CH}_3\text{CH}_2\text{C} \\ \diagdown \\ \text{O} \end{array}$ (1) ecf possible	1					
					1	1		3		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iv)		CH₃CH₂ CH₃ C=C CH₂CH₃ CH₃ (1) accept <i>E</i>- or <i>Z</i>- isomer 3,4-dimethylhex-3-ene (<i>E</i>- or <i>Z</i>- as structure given) (1) ecf possible from (a)(iii)II			1	2		
	(b)	(i)		award (1) for any of following • renewable source • does not use fossil fuels • method 2 gives two products (or biodiesel and propan-1,2,3-triol) • availability of raw materials	1			1		
		(ii)		award (1) each for any two of following • temperature conditions • pressure used • yield • rate • separation of products	2			2		2

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		award (1) for diagram and biodiesel labelled as top layer  award (1) for name of separating funnel	1		1	2		2
		(iv)		 partial / complete charges (1) curly arrows (1)		2		2		
				Question 9 total	7	4	4	15	0	5